COURSE INFORMATION

Faculty:	Computing	**Page:**	1 of 5
Program name:	Bachelor in Software Engineering with Honors		
Course code:	SCST1123	**Academic Session/Semester:**	2024-2025/1
Course name:	Mathematics for Software Engineer	**Pre/co requisite (course name and code, if applicable):**	Nil
Credit hours:	3		

Course synopsis	This course introduces students to the mathematical, computational, and linguistic foundations of software engineering. The mathematical topics covered include set theory, proof techniques, relations, functions, recurrence relations, counting methods, graph theory, and trees. Additionally, the course explores how formal language, and computational theories enhance the understanding of programming languages and their corresponding outputs—software. The focus will be on languages, grammars, and abstract machines, such as regular languages, context-free languages, regular grammars, context-free grammars, finite automata, pushdown automata, and Turing machines. By the end of the course, students should be able to apply set theory, relations, and functions to solve computer science problems; analyze and solve problems using recurrence relations and counting methods; apply graph theory and trees to real-world challenges; and utilize abstract machines to model electronic devices and problems.			
Course coordinator (if applicable)	Dr Nor Azizah Ali			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail @utm.my**
	Dr Nor Azizah Ali			nzah
	Dr Nor Haizan Mohamed Radzi			haizan
	Dr Mohamad Shukor Talib			shukor

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO	PLO (Code)	Taxonomies and generic skills	T&L methods	Assessment methods
CLO1	Analyse set theory, proof techniques, relations, functions, recurrence relation and counting methods to solve computer science problems.	PLO1 (KW)	C4	Lecture, active learning	A, Q, T, F
CLO2	Explain basic computing models and construct the equivalent abstract machines that accept the languages.	PLO2 (CG)	C3	Lecture, active learning	A, F
CLO3	Analyse relevant information regarding the theories and models in solving given problem.	PLO7 (NS)	C4	Lecture, active learning	A, Q, T, F, PR
CLO4	Suggest further improvement in the models for the solution to the problem.	PLO9 (PRS)	AD3	Project-Based learning	PR

Tu – Tutorial; T – Test; A – Assignment; PR – Project; F – Final Exam; Q- Quiz

Prepared by:	Certified by:
Name: Dr. Nor Azizah Ali	Name: Dr Sharin Hazlin binti Huspi
Signature: 	Signature: 
Date: 29 Sept. 2024	Date: 1 Oktober 2024

COURSE INFORMATION

School/Faculty:	Computing/Engineering	**Page:**	2 of 5
Program name:	Bachelor in Software Engineering with Honors		
Course code:	SCST1213	**Academic Session/Semester:**	2024-2025/1
Course name:	Mathematics for Software Engineer	**Pre/co requisite (course name and code, if applicable):**	Nil
Credit hours:	3		

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class activities and tutorials based on informal cooperative learning.
2.	Project-based learning.	Conducted through case study assignment. Tasks are given in sequential steps throughout the semester. Students in a group of 3 are given a case study and produce solution to the given problem. The report must comply to the case study specifications and be given in the form of written report.

Weekly Schedule:

Week 1 6-10 Oct. 2024	**1.0 SET THEORY** 1.1 Set and Subset 1.2 Operations on Sets
Week 2 13-17 Oct. 2024 **ASG#1**	**2.0 NUMBER THEORY** 2.1 The integer and Division 2.2 Primes and Greatest Common Division (GCD) 2.3 Euclid's GCD Algorithm
Week 3 20-24 Oct. 2024 **QUIZ #1**	**3.0 FUNDAMENTALS OF LOGIC** 3.1 Propositions, Conditional Propositions and Logical Equivalences 3.2 Quantifiers 3.3 Proof Techniques
Weeks 4 27-31 Oct. 2024 *Deepavali (31 Oct)-Thu	**4.0 RELATIONS & FUNCTIONS** 4.1 Relations • Digraph • Matrices of Relations • Equivalence and Partial Order Relations
Week 5 3-7 Nov. 2024 **ASG#2**	4.2 Functions • One-to-one, Onto, Bijection, Inverse functions • Composition 4.3 Recursive Algorithm 4.4 Recurrence Relation

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School/Faculty:	Computing/Engineering	**Page:**	3 of 5
Program name:	Bachelor in Software Engineering with Honors		
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Course name:	Mathematics for Software Engineer	**Pre/co requisite (course name and code, if applicable):**	Nil
Credit hours:	3		

Week 6 10-14 Nov. 2024	**5.0 COUNTING METHODS & COMBINATORICS** 5.1 Basic Principles 5.2 Permutations 5.3 Combination
Weeks 7 17-21 Nov. 2024 **Project – Briefing**	
Week 8 24-28 Nov. 2024	**MID-SEMESTER BREAK**
Weeks 9 1-6 Dec. 2024 **TEST (Mon, 2 Dec; 5pm – 7pm)**	**6.0 DISCRETE PROBABILITY** 6.1 Discrete Probability Theory 6.2 Bayes' Theorem
Weeks 10 8-12 Dec. 2024 **ASG#3**	**7.0 GRAPH THEORY** 7.1 Graph Definition and Notations 7.2 Representation of Graphs 7.3 Isomorphism of Graphs 7.4 Path and Cycles, Euler, Hamiltonian Cycles 7.5 Dijkstra's Shortest Path Algorithm 7.6 Trees and Spanning Tree
Week 11 15-19 Dec. 2024 **QUIZ #2**	
Week 12 22-26 Dec. 2024 *Christmas (25 Dec)-Wed	**8.0 LANGUAGE** 8.1 Language and String 8.2 Regular Language and Regular Expressions
Week 13 29 Dec 2024 -2 Jan. 2025 **ASG#4**	**9.0 FINITE AUTOMATA** 9.1 Finite Automata and Regular Expressions 9.2 Deterministic Finite Automata (DFA) 9.3 Non-Deterministic Finite Automata (NFA)
Week 14 5-9 Jan. 2025	**10. PUSH DOWN AUTOMATA (PDA)** 10.1 Designing Push Down Automata 10.2 Context Free Language
Week 15 12-16 Jan. 2025 **Project - Presentation**	**11.0 TURING MACHINE** Intro to Standard Turing Machine

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School/Faculty:	Computing/Engineering	**Page:**	4 of 5
Program name:	Bachelor in Software Engineering with Honors		
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Course name:	Mathematics for Software Engineer	**Pre/co requisite (course name and code, if applicable):**	Nil
Credit hours:	3		

Transferable skills (generic skills learned in course of study which can be useful and utilized in other settings):

Thinking skills
Problem solving

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline	**Teaching and Learning Activities**						**TOTAL SLT**
	Guided Learning Face to Face (F2F)				**Guided Learning Non Face to Face (NF2F)**	**Independent Learning Non F2F**	
CLO	**L**	**T**	**P**	**O**			
CLO 1	9	3	-	-	-	20	32
CLO 2	10	1	-	-	-	17	28
CLO 3	9	1		-	-	15	25
CLO4	-	-	-	9	4	9.5	22.5
Grand Total SLT	28	5	-	9	4	61.5	**107.5**

Continuous Assessment		**PLO**	**Percentage**		**Total SLT**
1	Assignment (4)	**PLO** 1 (KW)	5	20%	4h
		PLO 2 (CG)	5		
		PLO 7 (NS)	10		
2	Quiz (2)	**PLO** 1 (KW)	5	10%	1h
		PLO 7 (NS)	5		
3	Test	**PLO** 1 (KW)	13	20%	2.5h
		PLO 7 (NS)	7		
4	Project	**PLO** 7 (NS)	5	10%	2h
		PLO9 (PRS)	5		As in CLO3 (1h)
Summative Assessment		**PLO**	**Percentage**		**Total SLT**
1	Final Exam	**PLO 1** (KW)	10	40%	
		PLO 2 (CG)	4		3h
		PLO 7 (NS)	26		
Grand Total			100		120h

L: Lecture, T: Tutorial, P: Practical, O: Others

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

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School/Faculty:	Computing/Engineering	**Page:**	5 of 5
Program name:	Bachelor in Software Engineering with Honors		
Course code:	SCST1213	**Academic Session/Semester:**	2023/24/2
Course name:	Mathematics for Software Engineer	**Pre/co requisite (course name and code, if applicable):**	Nil
Credit hours:	3		

Learning resources:

Main references:

i. Johnsonbaugh, R. *Discrete Mathematics*, 8th ed. Pearson Prentice Hall, 2017.
ii. Thomas Sudkamp, Language and machine, Pearson Int. Edition, Third Edition, 2006.

Additional references

i. Malik, D.S. and Sen, M.K. *Discrete Mathematical: Theory and Applications*. Cengage Learning, 2012.
ii. Kenneth H. R., *Discrete Mathematical And Its Application",* 7th ed. Mc Graw Hill, 2012.
iii. Kolman, B., Busby, R.C.& Ross, S.C. *Discrete Mathematical Structure*, 4th .Ed.Prentice Hall, New Jercy, 1996.
iv. John C. Martin, Introduction to Languages and the Theory of Computation, Fourth Edition, 2011.
v. Michael Sipser, Introduction to the Theory of Computation, Cengage Learning, Third Edition, 2013.
vi. Elaine Rich. Automata, Computability and Complexity. Pearson International Edition. Pearson Prentice Hall. 2009.

Academic honesty and plagiarism: *(Below is just a sample):*

Any form of plagiarisms is NOT ALLOWED. Students who are caught cheating during exams may FAIL the course. Students who copied other student's assignment/exercise will get zero mark.

Other additional information (Course policy, any specific instruction etc.):

1. Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.
2. Students are required to behave and follow the University's dressing regulation and etiquette all the time.
3. Exercises and tutorial will be given in class, and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
4. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.
5. Make up exam will not be given, except to students who are sick and submit medical certificate which is confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.

Disclaimer:

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